INCH-POUND

A-A-52418
11 January 1993

COMMERCIAL ITEM DESCRIPTION

LIGHT, WARNING, VEHICULAR: ROTATING UNIT, 14 AND 28 VOLT DC

The General Services Administration has authorized the use of this commercial item description (CID) as a replacement for MIL-L-45325E and MS51317D, which are canceled.

1.0 Abstract. This CID covers 14- and 28-volt (V) direct current (DC) rotating warning lights for use on emergency, security, utility, or maintenance vehicles with 12- or 24- V DC electrical systems. The rotating warning light is referred to herein as "warning lights".

1.1 Classification. Warning lights shall be of the following types and classes, as specified (see 2.2 and 5.2):

Type I - 14 V DC
Type II - 28 V DC
Class 1 - Yellow/Amber dome.
Class 2 - Blue dome.
Class 3 - Red dome.

2.0 Salient characteristics.

2.1 Materials. Materials shall be as specified herein and in referenced documents. Materials shall be free from defects which adversely affect the performance of the finished product. The use of recovered materials made in compliance with regulatory requirements is acceptable providing that all requirements of this CID are met (see 5.5).

2.1.1 Dome material. The plastic used in fabricating domes shall meet the color and performance requirements of Society of Automotive Engineers (SAE) standard J576 and SAE J578.

Beneficial comments, recommendations, additions, deletions clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000.

AMSC N/A FSC 6220

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2.1.2 Skirt material. The skirt material shall be metal or plastic and shall meet the requirements specified herein.

2.1.3 Dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals which would encourage galvanic action shall be avoided. Separation of dissimilar metals shall be accomplished by providing insulation between mating surfaces.

2.2 Design and construction. Warning lights shall be designed and constructed in accordance with figure 1.

2.2.1 Lamps. Warning lights shall utilize two or more sealed beam lamps and shall be wired such that failure of one lamp shall not affect the other functions of the warning light. Mechanism for rotating lamps shall be shielded to prevent interference with wiring.

2.2.2 Soldering. Soldering shall be in accordance with MIL-STD-2000.

2.2.3 Mounting hardware. Mounting hardware, including the skirt, shall not be part of the electrical circuit.

2.2.4 Dome. Dome shall be designed to allow ease of removal and re-installation for periodic replacement of lamp(s)

2.2.5 Wires. Lead-in wires shall meet the requirements of SAE J1128 No. 16, type HDT for low tension, heavy duty, single conductor, unshielded cable. The wire shall terminate with connector MS27144-2 and MS27142-3 (one each).

2.2.6 Wire routing. Wires shall be routed through a hole in the skirt and shall be protected from sharp edges by a rubber or plastic grommet sealed with silicone RTV. Wires shall extend a minimum of 12 inches below the base of the skirt.

2.3 Performance.

2.3.1 Operation. When supplied with the voltage specified in table I for 35 ± 5 minutes, warning lights shall meet the flashing rate and current requirements of table I.

TABLE I. Operation requirements.

Type	Voltage (V dc)	Flashing rate (flashes/minute)	Maximum direct current (amperes)
I	14 ± 0.5	90 ± 30	10.5
II	28 ± 0.5	90 ± 30	5.5

2.3.2 Candlepower. The warning lights shall meet the photometric requirements of SAE J845 for the classes and colors specified in table II when tested in accordance with SAE J575.

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TABLE II. Photometric requirements.

Class	Class per SAE J845	Color
1	1	Yellow
2	2	Signal Blue
3	1	Red

2.3.3 Endurance. Warning lights shall exhibit no electrical or mechanical malfunction, except lamp failure, after being operated for 500 hours of on-time with a cycle of 25 minutes on, 5 minutes off. If the lamp burns out, the lamp shall be replaced and the test continued. Subsequently warning lights shall meet the operation requirements of 2.3.1, except the test duration shall be five minutes.

2.4 Environmental.

2.4.1 Extreme temperature. Warning lights shall meet the requirements of both the high temperature and low temperature tests of SAE J845.

2.4.2 Vibration. Warning lights shall show no evidence of disintegration of component materials or assemblies and shall meet the operation requirements of 2.3.1 for 5 minutes after being subjected to a simple harmonic motion of 0.076 cm (0.152 cm maximum total excursion), the frequency being varied uniformly between 10 and 30 Hertz (Hz). The entire frequency range, 10 to 30 Hz and back to 10 Hz, shall be traversed in approximately one minute. This motion shall be applied for 1 hour in each of three mutually perpendicular axis (total of 3 hours). Warning lights shall be operated for the last 10 minutes of each hour for each axis.

2.4.3 Shock. Warning lights shall show no evidence of disintegration of component materials or assemblies after being subjected to 18 induced shocks of sawtooth waveform with an amplitude of 25g and a duration of 11 milliseconds (ms). Three shocks shall be applied in both directions of each of three mutually perpendicular axes. Warning lights shall be operated while the shocks are applied. Subsequently, warning lights shall meet the operation requirements of 2.3.1, except the test duration shall be five minutes.

2.4.4 Salt fog. Warning lights shall show no evidence of interior or exterior corrosion after exposure to a 5% salt fog at 95°F for 96 hours. Warning lights shall be allowed to dry for 48 hours in an ambient environment prior to examination. The 5% salt solution shall have a pH of 6.5 to 7.2. Subsequently, warning lights shall meet the operation requirements of 2.3.1, except the test duration shall be five minutes.

2.4.5 Water spray. Warning lights shall meet the requirements of SAE J575 for water spray. Subsequently, warning lights shall meet the operation requirements of 2.3.1, except the test duration shall be five minutes.

2.4.6 Electromagnetic interference.

2.4.6.1 Conducted emissions. Electromagnetic emissions shall not appear on DC leads and interconnecting control leads in excess of the limits in figures 2 and 3 for narrowband and broadband emissions, respectively.

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2.4.6.2 Radiated emissions. Narrowband electric field emissions shall not be radiated in excess of the limits in figure 4 when measured at a distance of one meter. Broadband electric field from all equipment and subsystems, including radiated switching transients from (1) automatic cycling of electronic or electrical switching circuitry, or (2) manual switching shall not be radiated in excess of the limits in figure 5 when measured at a distance of one meter. Above 30 MHz the limits for both narrowband and broadband electric field radiated emissions shall be met for both horizontally and vertically polarized waves.

2.4.7 Weather resistance. Warning lights shall be exposed to 56 cycles of the solar radiation level in figure 6 (where one 24 hour duration of figure 6 is one cycle). No deviation of spectral distribution for thermal effects shall be allowed. Warning lights shall be mounted as in the vehicle with actual attaching hardware. Subsequently, warning lights shall meet the requirements of 2.2.3 and 2.3.2.

2.4.8 Heat aging. Warning lights shall show no evidence of cracking, warping, or crazing after exposure to ambient temperatures of not less than 200°F for not less than 240 hours. Subsequently, warning lights shall meet the requirements of 2.1.1 and 2.3.2.

2.4.9 Dust. Warning lights shall meet the requirements of SAE J575 paragraph 4.3 for dust exposure. Subsequently, warning lights shall meet the operation requirements of 2.3.1, except the test duration shall be five minutes.

2.5 Identification and markings. Unless otherwise specified (see 5.2), identification and marking shall be permanent and legible, and shall include as a minimum, the manufacturer's CAGE code number, the warning light voltage rating, and CID PIN (see 5.4).

2.6 Installation instructions. Each warning light shall be furnished with complete instructions covering the installation of the warning light. A wiring diagram showing all of the electrical connections to be made shall be included in the installation instructions.

2.7 Workmanship. The workmanship shall be such quality as to assure that warning lights furnished under this CID are free of defects that compromise, limit, or reduce performance or intended use. Warning lights shall be free of burrs, corrosion, scratches, sharp edges, loose components, chips, and unnecessary oil or foreign materials.

3.0 Quality assurance provisions.

3.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

3.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this CID and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.0 Preservation, packaging, packing, labeling, and marking. Preservation, packaging, packing, labeling, and marking for the desired level shall be as specified in the contract (see 5.2).

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5.0 Notes.

(This section contains information of a general or explanatory nature t may be helpful but is not mandatory.)

5.1 Addresses for obtaining copies of referenced documents.

5.1.1 Government specifications and standards. MIL-STD-2000 "Standard Requirement for Soldered Electrical and Electronic Assemblies" is available from the Navy Publications and Printing Service Office, Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia PA 19111-5094.

5.1.2 Non-Government publications. SAE J575 "Tests for Motor Vehicle Lighting Devices and Components", SAE J576 "Plastic Materials for Use in Optical Parts Such as Lenses and Reflectors of Motor Vehicle Lighting Devices", SAE J578 "Color Specification for Electric Signal Lighting Devices", and SAE J845 "360 Degree Warning Lamp for Authorized Emergency, Maintenance and Service Vehicles" are available from Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.

5.2 Ordering data. Acquisition documents must specify the following:

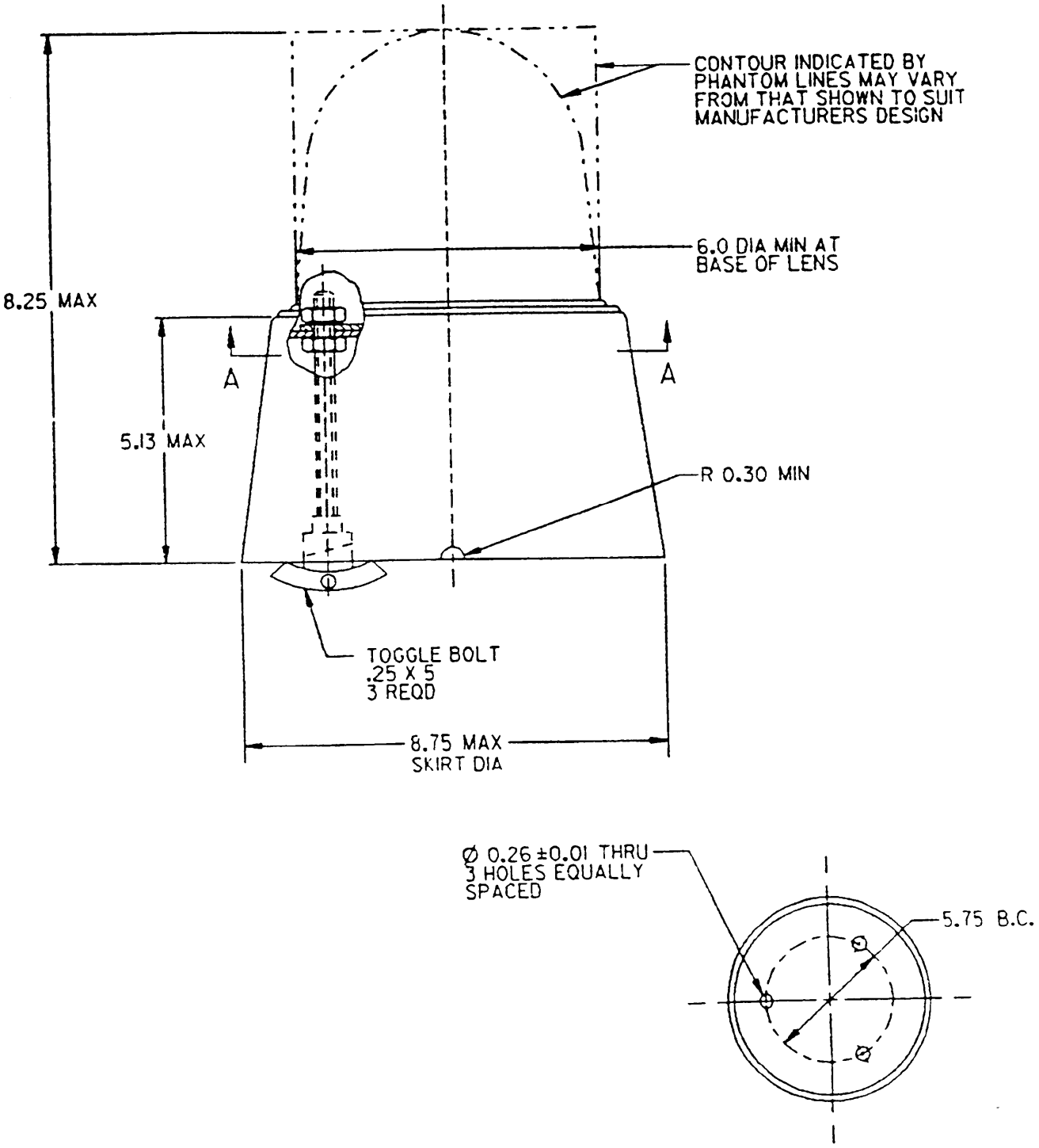
a. Title, number, and date of this CID.
b. Issue of Department of Defense Index of Specifications and Standards (DODISS) to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 5.1).
c. Type and class of warning light and PIN (see 1.1 and 5.3).
d. Marking if other than as specified (see 2.5).
e. Level of preservation, packaging, labeling, and marking (see 4.0).

5.3 Part or identification number (PIN). The PIN to be used for warning lights acquired to this CID is as follows:

A 52418 - XX - X

- Class (color of dome):
  - B - Blue
  - R - Red
  - Y - Yellow/Amber
- Type (voltage rating):
  - 14 - 14 V dc
  - 28 - 28 V dc
- CID number.
- Commercial item designator.

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NOTE: ALL DIMENSIONS IN INCHES

REDUCED VIEW SHOWING TOGGLE BOLT HOLES IN BASES PLATE

FIGURE 1. Warning light.

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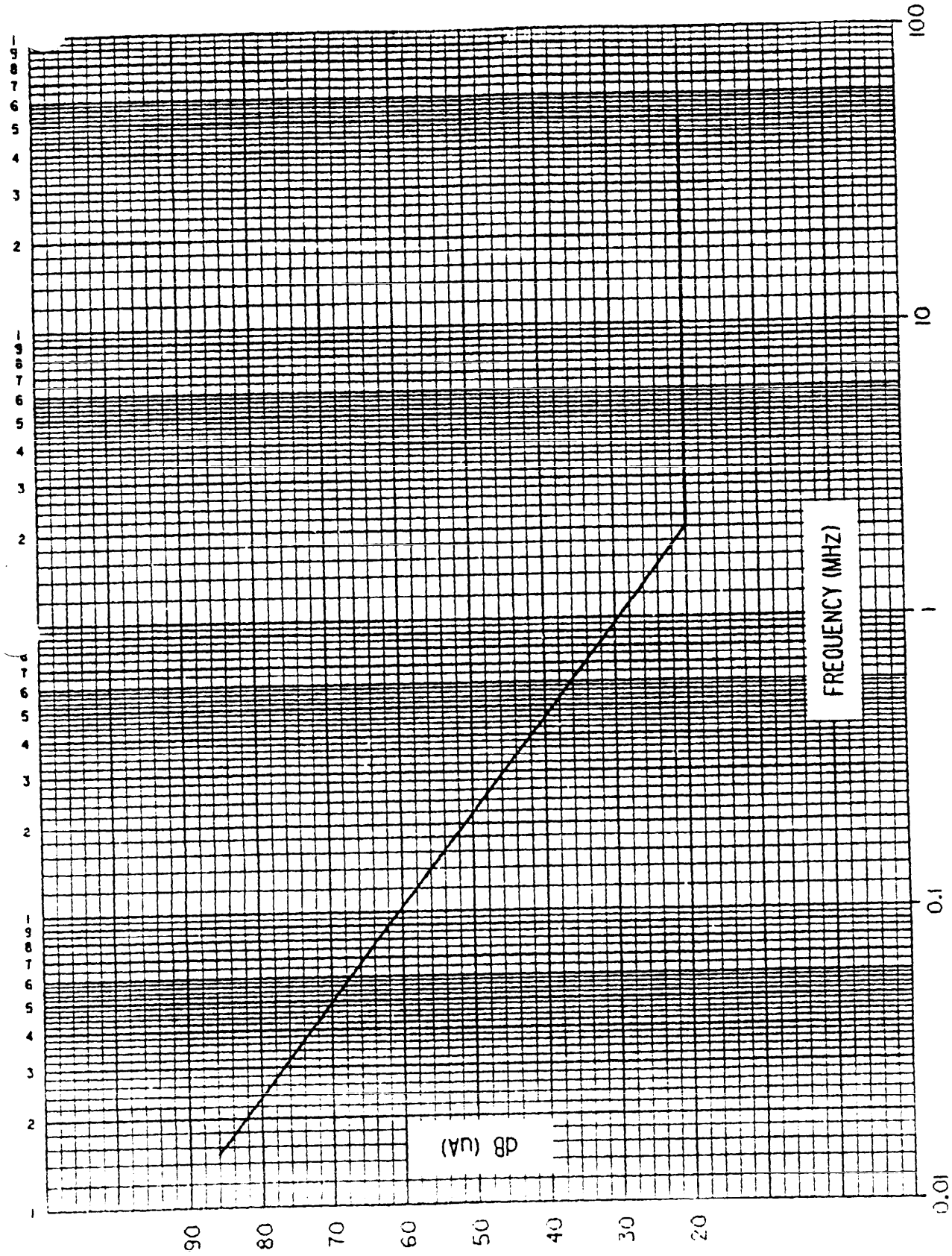


FIGURE 2. Limit for narrowband, conducted emissions - DC leads and interconnecting leads.

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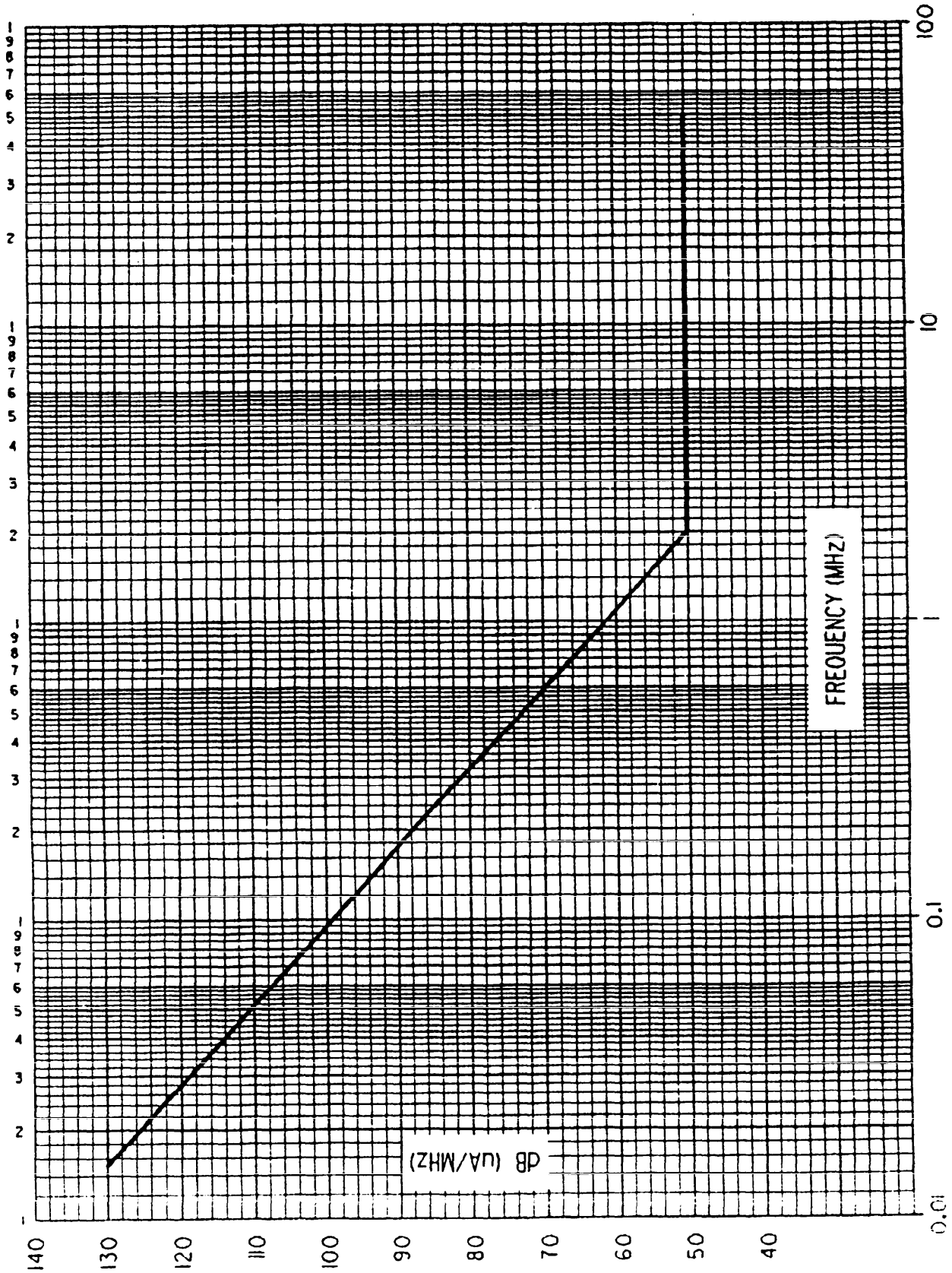


FIGURE 3. Limit for broadband, conducted emissions - DC leads and interconnecting leads.

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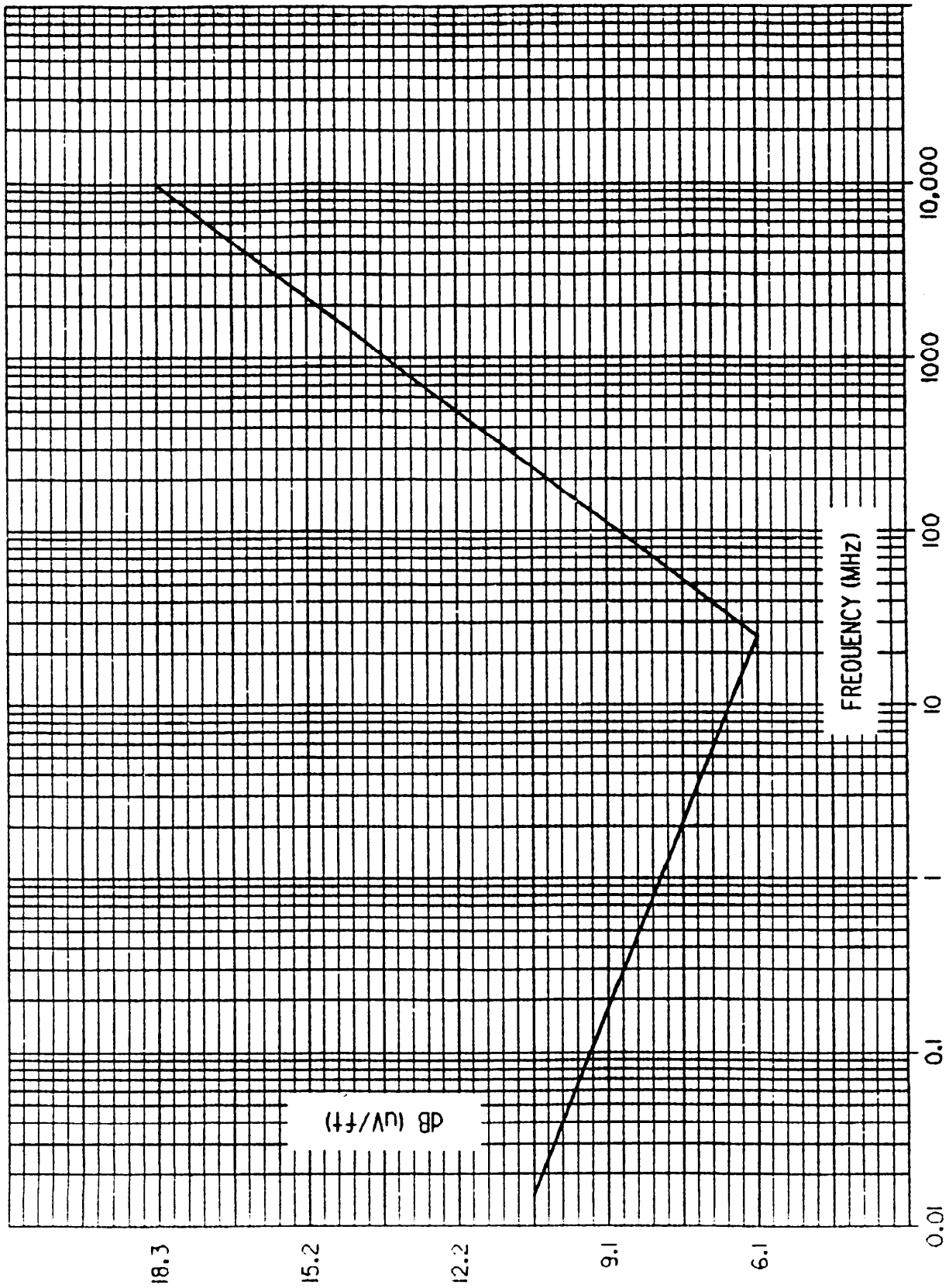


FIGURE 4. Limit for narrowband, radiated emissions.

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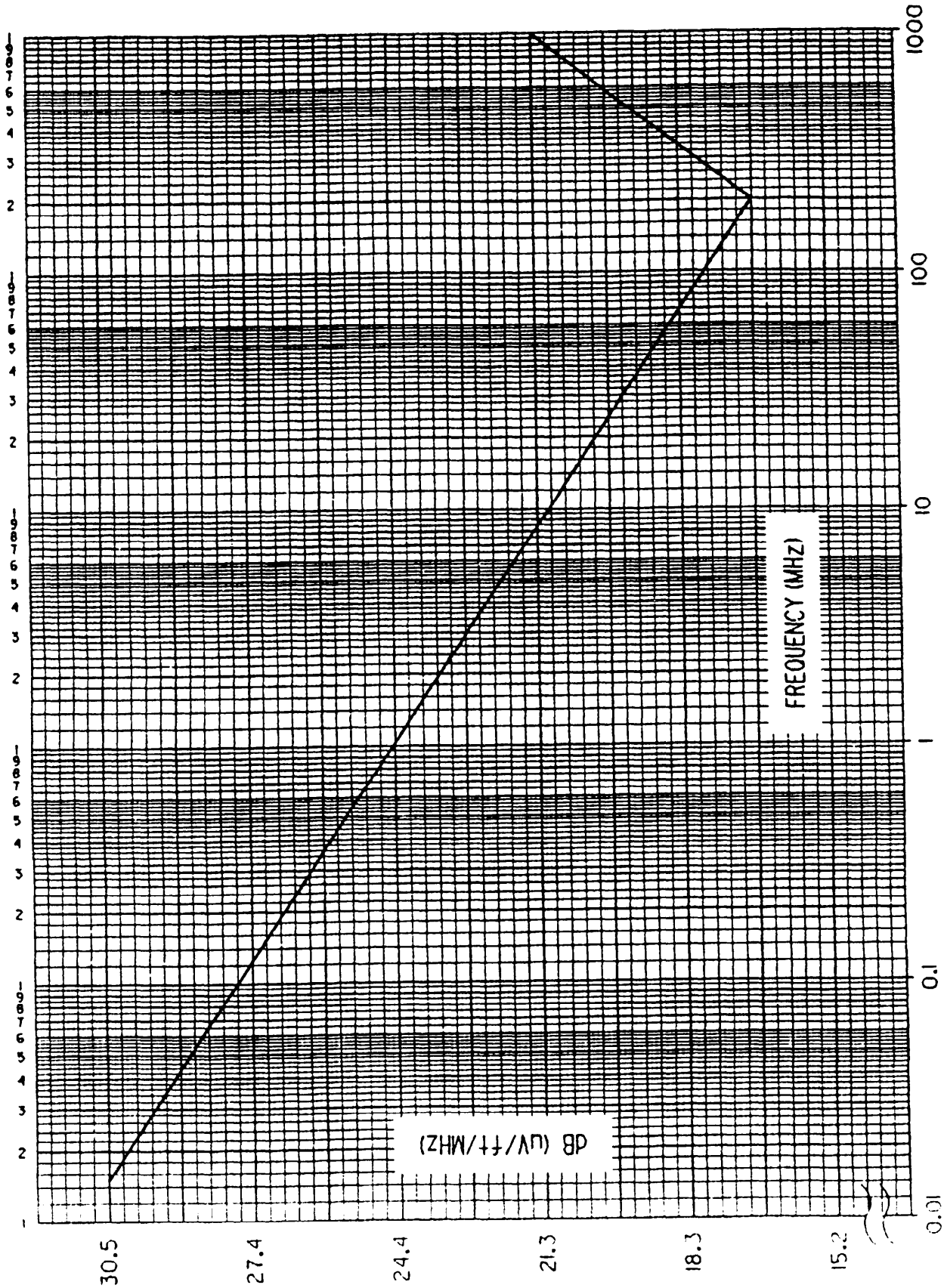


FIGURE 5. Limit for broadband, radiated emissions.

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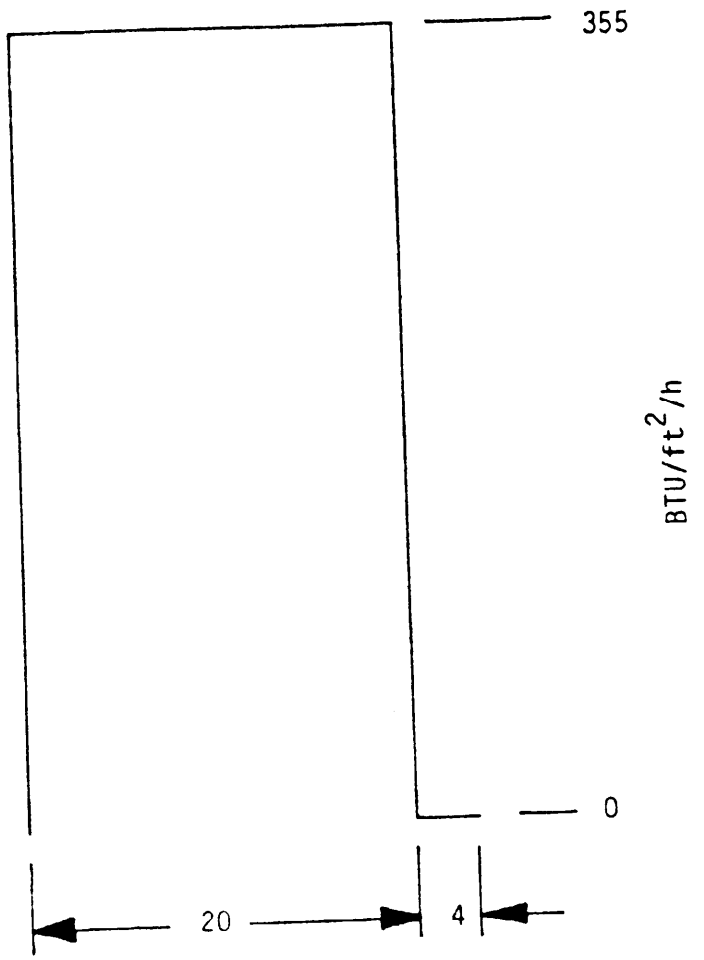


FIGURE 5. Weather resistance test cycle duration.

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5.4 Cross-reference. Warning lights procured to this CID are substitutable/ interchangeable with warning lights procured to MS51317D as shown in table IV:

TABLE IV. CID PIN to MS dash no. cross-reference,

CID PIN	Superseded MS part no.
A 52418-14-Y	MS51317-4
A 52418-28-Y	MS51317-3
A 52418-14-B	MS51317-6
A 52418-28-B	MS51317-5
A 52418-14-R	MS51317-2
A 52418-28-R	MS51317-1

5.5 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

5.6 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of American Society for Testing and Materials (ASTM) standard E380, and all other requirements of this CID are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Custodians:
Army - AT
Navy - YD
Air Force - 99

Review activities:
Army - ME
DLA - GS
Air Force - 82

User activities:
Navy - MC, SH

Preparing activity:
Army - AT

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